Assignment 2 : Assignments on Operators

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Q 1.

1. **Problem Statement:** Program to find maximum of three numbers

   Code:

```python
num1=int(input("Enter the first number: "))
num2=int(input("Enter the second number: "))
num3=int(input("Enter the third number: "))

if num1>num2 and num1>num3:
    print("The largest number is: ",num1)
elif num2>num1 and num2>num3:
    printf("The largest number is: ",num2)
else:
    print("The largest number is: ",num3)

    print("The maximum number is:",max(num1,num2,num3))
```

   Output:

```
PS D:\Desktop\Assignment & Notes\Python> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python313/python.exe "d:/Desktop/Assignment & Notes/Python/1.py"
Enter the first number: 5
Enter the second number: 2
Enter the third number: 8
The largest number is:  8
The maximum number is: 8
PS D:\Desktop\Assignment & Notes\Python>
```

Conclusion:

Thus by taking input from user of three numbers and using conditional statement if, elif , else we get maximum of three numbers

Q 2.

Problem Statement: Program to check if a year is leap

Code:

```python
year = int(input("Enter a year: "))

if (year % 4 == 0):
    print(f"{year} is a leap year")
else:
    print(f"{year} is not a leap year")
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Desktop\Assignment & Notes\Python> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python313/python.exe "
d:/Desktop/Assignment & Notes/Python/2.py"
Enter a year: 2026
2026 is not a leap year
PS D:\Desktop\Assignment & Notes\Python> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python313/python.exe "
d:/Desktop/Assignment & Notes/Python/2.py"
Enter a year: 2024
2024 is a leap year
PS D:\Desktop\Assignment & Notes\Python>
```

Conclusion:

As leap year come after every 4 years it must be divisible by 4 if not the year is not leap year.

Q 3.

Problem Statement: Program to check if a date is valid

Code:

```python
day = int(input("Enter day (1-31): "))
month = int(input("Enter month (1-12): "))
year = int(input("Enter year: "))

if 1 <= day <= 31 and 1 <= month <= 12 and year > 0:
    print(f"{day}/{month}/{year} is valid")
else:
    print(f"{day}/{month}/{year} is invalid")
```

Output:

```
PS D:\Desktop\Assignment & Notes\Python> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python313/python.exe "d:/Desktop/Assignment & Notes/Python/3.py"
Enter day (1-31): 32
Enter month (1-12): 20
Enter year: 2026
32/20/2026 is invalid
PS D:\Desktop\Assignment & Notes\Python> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python313/python.exe "d:/Desktop/Assignment & Notes/Python/3.py"
Enter day (1-31): 12
Enter month (1-12): 12
Enter year: 2026
12/12/2026 is valid
```

Conclusion:

Taking input day, month, year from user and applying if-else condition we get the output whether the input date is valid or not.

Q 4.

1. **Problem Statement:** Program to find the roots of a quadratic equation

```python
import math
# program to find roots of quadratic equation

a=float(input("Enter coefficient a:"))
b=float(input("Enter coefficient b:"))
c=float(input("Enter coefficient c:"))
d=b**2-4*a*c

if d>0:
    root1=(-b+math.sqrt(d))/(2*a)
    root2=(-b-math.sqrt(d))/(2*a)
    print(f"root1={root1} and root2={root2}")
elif d==0:
    root=-b/(2*a)
    print(f"root={root}")
else:
    print("Roots are imaginary")
```

Output:

```
PS D:\Desktop\Assignment & Notes\Python> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python313/python.exe "d:/Desktop/Assignment & Notes/Python/4.py"
Enter coefficient a:5
Enter coefficient b:3
Enter coefficient c:1
Roots are imaginary
PS D:\Desktop\Assignment & Notes\Python>
```

Conclusion:

Taking input of coefficients of a, b, c from user and applying the formula of general solutions of quadratic equations we get the output

Q 5.

1. **Problem Statement:** Given a string. Delete from it all the characters whose indices are divisible by 3**.**

   Code:

```python
string=input("Enter a string:")
result = ''.join([string[i] for i in range(len(string)) if i%3 !=0])
print(result)
```

   Output:

```
PS D:\Desktop\Assignment & Notes\Python> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python313/python.exe "d:/Desktop/Assignment & Notes/Python/5.py"
Enter a string:my name is rajesh
y am i rjeh
PS D:\Desktop\Assignment & Notes\Python>
```

   Conclusion:

   This program removes characters from a string at indices divisible by 3 (positions 0, 3, 6, 9, ...)

Q 6.

Problem Statement: Given a sequence of integer numbers ending with the number 0. Determine the length of the widest fragment where all the elements are equal to each other.

Code:

```python
# Find longest fragment of equal elements

nums = []
while True:
    n = int(input("Enter number (0 to end): "))
    if n == 0:
        break
    nums.append(n)

max_len = 1
current_len = 1

for i in range(1, len(nums)):
    if nums[i] == nums[i-1]:
        current_len += 1
    else:
        max_len = max(max_len, current_len)
        current_len = 1

max_len = max(max_len, current_len)
print(f"Longest fragment length: {max_len}")
```

Output:

```
PS D:\Desktop\Assignment & Notes\Python> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python313/python.exe "d:/Desktop/Assignment & Notes/Python/6.py"
Enter number (0 to end): 2
Enter number (0 to end): 2
Enter number (0 to end): 2
Enter number (0 to end): 3
Enter number (0 to end): 3
Enter number (0 to end): 3
Enter number (0 to end): 4
Enter number (0 to end): 0
Longest fragment length: 3
PS D:\Desktop\Assignment & Notes\Python>
```

Conclusion:

This program finds the length of the longest consecutive sequence of equal numbers from user input (ends with 0).

Q 7.

Problem Statement: In bowling, the player starts with 10 pins at the far end of a lane. The object is to knock all the pins down. For this exercise, the number of pins and balls will vary. Given the number of pins N and then the number of balls K to be rolled, followed by K pairs of numbers (one for each

ball rolled), determine which pins remain standing after all the balls have been rolled. The balls are numbered from 1 to N (inclusive) for this situation. The subsequent number pairs, one for each K represent the start to stop (inclusive) positions of the pins that were knocked down with each role. Print a sequence of N characters, where "I" represents a pin left standing and "." represents a pin knocked down.

Code:

```python
# Bowling pins problem

n = int(input("Enter number of pins: "))
k = int(input("Enter number of balls: "))

pins = ['I'] * n

for _ in range(k):
    start, stop = map(int, input("Enter range (start stop): ").split())
    for i in range(start - 1, stop):
        pins[i] = '.'

print(''.join(pins))
```

Output:

```
PS D:\Desktop\Assignment & Notes\Python> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python313/python.exe "d:/Desktop/Assignment & Notes/Python/7.py"
Enter number of pins: 5
Enter number of balls: 2
Enter range (start stop): 1 3
Enter range (start stop): 5 5
...I.
PS D:\Desktop\Assignment & Notes\Python>
```

Conclusion:

This program simulates a bowling game, displaying which pins remain standing ('I') and which are knocked down ('.') after all balls are rolled.

Q 8.

Problem Statement: A timestamp is three numbers: a number of hours, minutes and seconds. Given two timestamps, calculate how many seconds are between them. The moment of the first timestamp occurred before the moment of the second timestamp.

Code:

```python
# First timestamp
h1 = int(input("Enter first timestamp hours: "))
m1 = int(input("Enter first timestamp minutes: "))
s1 = int(input("Enter first timestamp seconds: "))

# Second timestamp
h2 = int(input("Enter second timestamp hours: "))
m2 = int(input("Enter second timestamp minutes: "))
s2 = int(input("Enter second timestamp seconds: "))

# Convert to total seconds
time1 = h1 * 3600 + m1 * 60 + s1
time2 = h2 * 3600 + m2 * 60 + s2

difference = time2 - time1

print(f"Difference: {difference} seconds")
```

Output:

```
PS D:\Desktop\Assignment & Notes\Python> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python313/python.exe "d:/Desktop/Assignment & Notes/Python/8.py"
Enter first timestamp hours: 5
Enter first timestamp minutes: 36
Enter first timestamp seconds: 12
Enter second timestamp hours: 6
Enter second timestamp minutes: 25
Enter second timestamp seconds: 15
Difference: 2943 seconds
PS D:\Desktop\Assignment & Notes\Python>
```

Conclusion:

This program calculates the total number of seconds between two given timestamps.